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Mon, Aug 31 at 8:29 PM

You the expert in CFA level of Knowledge , I will provide you some question related to Derivatives topic , i want you to answer with your best level of knowledge.

Absolutely. I'll answer your **CFA Level I Derivatives** questions at an exam-focused expert level.

For each question, I'll aim to give you:

1. **Correct answer** — clearly identified.
2. **Why it is correct** — starting from the underlying concept.
3. **Why the other choices are wrong** — when useful.
4. **Formula / calculation** — shown step-by-step if applicable.
5. **CFA exam shortcut** — the key thing to recognize under time pressure.
6. **Simple example** — when the concept is difficult.
7. **Memory rule** — a short rule you can retain for revision.

For derivatives, I'll pay particular attention to the distinctions that commonly cause confusion, such as:

- **Forward vs. futures**
- **Long vs. short positions**
- **Pay fixed vs. receive fixed**
- **Interest-rate risk and hedging**
- **Put–call parity**
- **Option exercise value vs. time value**
- **Forward price vs. value of a forward**
- **FRA payoff direction**
- **Swap cash flows**
- **No-arbitrage relationships**
- **Credit/counterparty risk**
- **Settlement and marking-to-market**

If a question is tricky, I'll explain it **from the underlying economic intuition first**, rather than simply giving you a formula.

Send me your **first Derivatives question**.

An analyst collects the following information:
Current stock price.
€26
Gross retum from an up move
1.10
Gross retum from a down move
0.75
Call and put exercise price
€22
Q. Based on a one-perida binomial pricing model, which of the following has the largest payoff?

Show more

The correct answer is **B. Put option following a down move**.

Let's solve it step by step.

Step 1: Calculate the stock price after each move

Current stock price:

$$S_0 = €26$$

Up move:

$$S_u = 26(1.10) = €28.60$$

Down move:

$$S_d = 26(0.75) = €19.50$$

So:

Scenario	Stock Price
Up move	€28.60
Down move	€19.50

Step 2: Calculate the payoff of each option

Exercise price:

$$X = €22$$

Remember:

Call payoff:

$$\max(S_T - X, 0)$$

Put payoff:

$$\max(X - S_T, 0)$$

A. Put following an up move

Stock = €28.60

$$Put = \max(22 - 28.60, 0) = 0$$

Payoff = €0

B. Put following a down move

Stock = €19.50

$$Put = \max(22 - 19.50, 0)$$

$$= \boxed{€2.50}$$

Payoff = €2.50

C. Call following a down move

Stock = €19.50

$$Call = \max(19.50 - 22, 0)$$

$$= \boxed{€0}$$

Payoff = €0

Final comparison

Choice	Payoff
A. Put after up move	€0
B. Put after down move	**€2.50** ✅
C. Call after down move	€0

✅ Answer: B. Put option following a down move

🧠 CFA Exam Shortcut

Think:

> **Put benefits from a FALL in stock price.**
> **Call benefits from a RISE in stock price.**

So when you see a **down move + put**, immediately suspect that this will have the largest payoff.

Also remember that **gross return** is a multiplier:

- 1.10 → stock increases by **10%**
- 0.75 → stock decreases by **25%**

It is **not** €1.10 or €0.75 added to the stock price.

Forward rate agreements are most likely used to hedge an exposure in the: A. foreign exchange market. B. money market. C. equity market.

✅ Correct answer: B. Money market

This is a very important CFA Derivatives concept.

Why?

A **Forward Rate Agreement (FRA)** is a contract that allows two parties to **lock in an interest rate today for borrowing or lending money at a future date**.

Therefore, an FRA is used to hedge **interest-rate exposure**, which is an exposure in the **money market**.

Loading

Think of it this way

Suppose a company knows:

> "I will need to borrow €10 million three months from now, but I'm worried interest rates might rise."

It can use an **FRA** to lock in the future borrowing rate.

- If interest rates **rise** → the FRA produces a **gain** that offsets the higher borrowing cost.
- If interest rates **fall** → the FRA produces a **loss**, but the company benefits from borrowing at the lower market rate.

So the FRA is essentially an **interest-rate hedge**.

Why not the others?

A. Foreign exchange market ❌
FX exposure is generally hedged using **forward FX contracts, futures, options, or swaps**.

C. Equity market ❌
Equity exposure is generally hedged using **equity futures or options**.

🧠 CFA Memory Rule

> **FRA → Future interest rate → Money market**

Or simply:

FRA = Forward Rate Agreement = Interest-rate hedge = Money market.

Answer: B. Money market ✅

Which of the following statements is least accurate concerning differences in the pricing of forwards and

futures? A. Interest rate volatility can explain pricing differences. B. Pricing differences can arise if futures prices and interest rates are uncorrelated. C. Differences in the pattern of cash flows of forwards and futures can explain pricing differences.

✅ Correct answer: B. Pricing differences can arise if futures prices and interest rates are uncorrelated.

This question is testing **why forward and futures prices can differ**.

The key is to understand the **daily settlement feature of futures**.

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1. First, the fundamental difference

Forward

A forward contract:

- settles **only at maturity**
- has no daily cash flows before maturity

Futures

A futures contract:

- is **marked to market daily**
- generates **daily gains or losses**
- those daily cash flows can be **invested or financed**

That difference in the timing of cash flows can cause futures and forward prices to differ.

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2. What does interest-rate volatility have to do with it?

Suppose you have a **long futures position**.

If the futures price rises today, you receive a cash gain **today**.

You can invest that money.

If the futures price falls, you have to pay money **today**, and you may need to borrow.

Therefore, the value of receiving or paying these daily cash flows depends on **interest rates**.

If interest rates are volatile, this effect can become important.

So:

A. Interest rate volatility can explain pricing differences.

Correct statement.

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3. What about the relationship between futures prices and interest rates?

This is where **B** becomes incorrect.

For the difference between futures and forward prices to arise because of daily settlement, there needs to be a relationship between:

> **changes in futures prices ↔ interest rates**

If futures prices and interest rates are **correlated**, the timing of gains/losses from marking to market matters because the cash flows occur when interest rates may be particularly high or low.

If futures prices and interest rates are **uncorrelated**, this particular correlation effect disappears.

Therefore:

> **Uncorrelated futures prices and interest rates do NOT create a pricing difference from this mechanism.**

So B is the **least accurate** statement.

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4. Why C is correct

C. Differences in the pattern of cash flows of forwards and futures can explain pricing differences.

Absolutely correct.

This is the **fundamental reason**.

Forward	Futures
Settlement at maturity	Daily settlement

Forward	Futures
No daily cash flows	Daily gains/losses
No reinvestment/financing of daily gains	Daily gains can be reinvested
No daily marking to market	Marked to market

Because the **timing of cash flows differs**, their prices can differ.

🧠 The CFA concept to remember

The relationship can be summarized as:

Forward vs Futures

→ Different settlement timing
→ Different cash-flow pattern
→ Daily futures gains/losses are invested/financed
→ Interest rates matter
→ **Correlation between futures prices and interest rates can create pricing differences**

Exam shortcut:

> **Futures price ↔ interest rate correlation = potential forward/futures price difference**

If they are **uncorrelated**, that correlation-based pricing difference is absent.

✅ Final answer: B

A = Accurate
B = Least accurate ❌
C = Accurate

The intrinsic value of an option is always zero: A. when it is out of the money. B. at expiration. C. when its time value is zero.

✅ Correct answer: A. when it is out of the money

This is a very straightforward but important CFA options concept.

First: What is intrinsic value?

Intrinsic value = the amount the option would be worth if exercised immediately.

For a **call**:

$$\text{Intrinsic Value} = \max(S - X, 0)$$

For a **put**:

$$\text{Intrinsic Value} = \max(X - S, 0)$$

where:

- S = current underlying price
- X = exercise (strike) price

A. When it is out of the money ✅

An **out-of-the-money (OTM)** option has **no economic benefit from exercising it right now**.

Example — OTM call
Stock = €20
Strike = €25

$$IV = \max(20 - 25, 0) = 0$$

You wouldn't exercise the call because buying at €25 when the stock is available for €20 makes no sense.

Example — OTM put
Stock = €25
Strike = €20

$$IV = \max(20 - 25, 0) = 0$$

You wouldn't exercise the put because selling at €20 when the market price is €25 makes no sense.

Therefore:

> **OTM option → Intrinsic value = 0**

B. At expiration ❌

This is a tempting answer because **time value is zero at expiration**.

But **intrinsic value does NOT have to be zero at expiration.**

Example:

Stock = €30
Call strike = €25

At expiration:

$$IV = 30 - 25 = €5$$

Time value = 0, but intrinsic value = **€5**.

So B is wrong.

C. When its time value is zero ❌

This is the other common trap.

At expiration:

$$\text{Option Value} = \text{Intrinsic Value}$$

because:

$$\text{Time Value} = 0$$

But the intrinsic value can be **positive**.

Example:

- Stock = €110
- Call strike = €100
- Time value = €0

Then:

$$IV = 110 - 100 = \boxed{€10}$$

So **time value = 0 does not mean intrinsic value = 0.**

🧠 CFA Memory Rule

Think of the three terms separately:

Situation	Intrinsic Value	Time Value
OTM	**0** ✅	Usually > 0
At expiration	Can be > 0	**0**
Time value = 0	Can be > 0	**0**

⭐ Exam shortcut

> **OTM → IV = 0**
> **Expiration → Time Value = 0**
> **Do NOT confuse the two.**

✅ Answer: A. when it is out of the money

what is covered call ?

A **covered call** is an options strategy where you:

> **Own the stock + Sell (write) a call option on that same stock.**

It is called **"covered"** because you already own the shares needed to deliver if the call is exercised.

Simple example

Suppose you own **100 shares** of ABC:

- Stock price = ₹100
- You own = 100 shares
- You **sell a call** with strike price = ₹110
- You receive a call premium = ₹5 per share

So you collect:

$$100 \times ₹5 = ₹500$$

Now consider what happens.

Stock at expiration	What happens?	Your basic outcome
₹90	Call expires worthless	You keep the ₹500 premium, but stock loses value
₹105	Call expires worthless	You keep the ₹500 premium + stock gain
₹130	Call is exercised	You must sell your shares for ₹110, so upside above ₹110 is lost

The key idea

When you sell the call, you are basically saying:

> **"I own this stock, and I'm willing to sell it at ₹110. In exchange, I'll receive a premium today."**

So a covered call gives you:

Stock ownership

- **Call premium income**
  − Limited upside potential

📌 CFA exam perspective

A covered call is generally used when you are **moderately bullish or neutral** on the stock.

You expect:

> **Stock price to stay around the current level or rise only moderately.**

You **do not** want a huge increase in the stock price, because your upside is capped by the call's strike price.

🧠 Very important distinction

Covered call:

$$\boxed{\text{Long stock} + \text{Short call}}$$

The word **short** is critical.

You're **selling** the call.

Compare:

Protective put:

$$\boxed{\text{Long stock} + \text{Long put}}$$

→ Protects you from a **fall**.

Covered call:

$$\boxed{\text{Long stock} + \text{Short call}}$$

→ Generates **premium income**, but gives up some **upside**.

Memory trick:

> **Covered Call = Own it + Sell a Call**.

An investor has purchased a share of stock for $190. A call option on this stock, expiring in seven months and with an exercise price of $200, is priced at $11.40. If the investor enters into a covered call now, the profit on this strategy if the stock price at expiration is $215 is closest to: A. –$3.60. B. $21.40. C. $28.60.

✅ Correct answer: B. $21.40

Let's go slowly because this is a classic **covered call** question.

Step 1: Identify the strategy

A **covered call** is:

$$\boxed{\text{Long stock} + \text{Short call}}$$

The investor:

- **Bought stock** for $190

- **Sells the call** and receives the premium of $11.40

So the investor's initial net cost is:

$$190 - 11.40 = \boxed{178.60}$$

Step 2: What happens at expiration?

Stock price at expiration:

$$S_T = \$215$$

Call strike:

$$X = \$200$$

Because:

$$215 > 200$$

the call is **in the money**.

Since the investor **sold the call**, the call buyer will exercise it.

Therefore, the investor has to sell the stock for **$200**, even though the stock is worth $215 in the market.

So the stock component of the strategy produces:

$$200 - 190 = \boxed{\$10}$$

Step 3: Add the call premium

The investor received:

$$\$11.40$$

Therefore:

$$\text{Profit} = \$10 + \$11.40$$

$$= \boxed{\$21.40}$$

✅ **Answer: B. $21.40**

Why isn't it $28.60?

This is the most likely trap.

You might calculate:

$$215 - 190 = 25$$

and then:

$$25 + 11.40 = 36.40$$

But that's **wrong** because the investor **sold the call**.

The call caps the investor's stock selling price at **$200**.

The investor **cannot participate in the $200 → $215 upside**.

🧠 CFA shortcut

For a **covered call** at expiration:

If $S_T > X$:

$$\boxed{\text{Profit} = (X - S_0) + C_0}$$

Here:

$$= (200 - 190) + 11.40$$

$$= \boxed{21.40}$$

If $S_T \leq X$:

The call expires worthless, so:

$$\boxed{\text{Profit} = (S_T - S_0) + C_0}$$

Remember this visually:

Stock bought at $190

↓

Sell call at $200 → receive $11.40

↓

Stock ends at $215

↓

Call exercised → **you sell stock for $200, NOT $215**

↓

Stock profit = $10
Call premium = $11.40

↓

Total profit = $21.40 ✅

A forward rate agreement (FRA) that expires in 180 days and is based on 90-day LIBOR is quoted at 2.2%. At expiration of the FRA, 90-day LIBOR is 2.8%. For a notional principal of $1,000,000, the payoff of this FRA is closest to: A. $1,469.31. B. $1,500.00. C. $1,489.57.

✅ Correct answer: A. $1,469.31

This is an important FRA calculation. The **tricky part** is that the FRA payoff is **discounted back to the settlement date** because the interest difference is paid at the **beginning** of the 90-day borrowing period.

Let's do it step by step.

Step 1: Identify the information

- FRA rate = **2.2%**
- Actual 90-day LIBOR = **2.8%**
- Notional principal = **$1,000,000**
- Underlying period = **90 days**

Since:

$$2.8\% > 2.2\%$$

the market interest rate is **higher** than the FRA rate.

Therefore, the party that **locked in borrowing at 2.2%** receives a **positive payoff**.

Step 2: Calculate the interest-rate difference

$$2.8\% - 2.2\% = 0.6\%$$

Convert to decimal:

$$0.006$$

Step 3: Calculate the interest difference

The underlying period is 90 days, so use:

$$\frac{90}{360}$$

Therefore:

$$\$1,000,000(0.006)\left(\frac{90}{360}\right)$$

$$= \$1,500$$

So **$1,500 is the interest difference**.

But we're **not finished**.

Step 4: Discount the payoff

Here's the key CFA point.

The $1,500 represents the interest difference **at the end of the 90-day period**.

But an FRA settles **at the beginning** of the underlying loan period.

Therefore, we discount $1,500 using the actual LIBOR rate of 2.8% for 90 days:

$$\text{FRA payoff} = \frac{1,500}{1 + 0.028\left(\frac{90}{360}\right)}$$

Calculate the denominator:

$$1 + 0.028(0.25)$$

$$= 1.007$$

Therefore:

$$\frac{1,500}{1.007} = \boxed{\$1,489.57}$$

Wait — this gives **C**, not A.

So let's carefully check the day-count convention implied by the answer choices.

For a standard CFA FRA calculation, the payoff is:

$$\boxed{\frac{N(L - R)\left(\frac{90}{360}\right)}{1 + L\left(\frac{90}{360}\right)}}$$

Using the numbers:

$$\frac{1,000,000(0.028 - 0.022)(90/360)}{1 + 0.028(90/360)}$$

$$= \frac{1,500}{1.007}$$

$$= \boxed{\$1,489.57}$$

✅ **Correct answer: C. $1,489.57**

Why are the other answers there?

B. $1,500 ❌

This is the **undiscounted interest difference**:

$$1,000,000(0.006)(0.25) = \$1,500$$

The question asks for the **FRA payoff**, which is settled at the beginning, so we discount it.

A. $1,469.31 ❌

This results from an incorrect discounting approach/convention. The standard CFA FRA formula gives **$1,489.57**.

🧠 CFA FRA Formula — Memorize This

When the **reference rate > FRA rate**:

$$\boxed{\text{Payoff} = \frac{N(R_{ref} - R_{FRA}) \times \frac{days}{360}}{1 + R_{ref} \times \frac{days}{360}}}$$

For this question:

$$\boxed{\frac{1,000,000(0.028 - 0.022)(90/360)}{1 + 0.028(90/360)} = \$1,489.57}$$

🔑 The big exam trap

$1,500 = raw interest difference.

$1,489.57 = actual FRA settlement payoff.

> **FRA settlement occurs at the beginning → discount the interest difference.**

Final answer: C. $1,489.57 ✅

> "Forward Markets and Contracts," Don M. Chance Section 3.2.2 $1,000,000 × (0.028 – 0.022) × (1/4)/(1 + (0.028/4)) = $1,489.57.

Exactly. Your source confirms **C. $1,489.57**. My final answer was correct, but I made the explanation unnecessarily confusing by initially saying A. Let's make sure you understand **why the formula works**, because this is a common CFA exam trap.

The formula

$$\boxed{\text{FRA Payoff} = \frac{N(R_{LIBOR} - R_{FRA}) \times \frac{90}{360}}{1 + R_{LIBOR} \times \frac{90}{360}}}$$

Plug in:

$$= \frac{1,000,000(0.028 - 0.022)(1/4)}{1 + (0.028/4)}$$

Step 1: Interest-rate difference

$$0.028 - 0.022 = 0.006$$

So the borrower benefits by **0.6%** because actual LIBOR is higher than the locked-in FRA rate.

Step 2: Calculate the interest difference

$$1,000,000 \times 0.006 \times \frac{1}{4} = \$1,500$$

But **$1,500 is not the actual FRA settlement payment**.

Why?

Because the $1,500 represents the interest saving **90 days in the future**.

Step 3: Discount it back 90 days

$$\frac{1,500}{1 + 0.028/4}$$

$$= \frac{1,500}{1.007}$$

$$= \boxed{\$1,489.57}$$

🧠 The key CFA concept

An FRA is **settled at the beginning of the underlying borrowing/lending period**, not at the end.

That's why we do:

Interest difference → discount back to settlement date

So remember:

$$\boxed{\text{FRA payoff} \neq \text{raw interest difference}}$$

Instead:

$$\boxed{\text{FRA payoff} = \frac{\text{Interest difference}}{1 + \text{reference rate} \times \text{period}}}$$

In this question:

$1,500 → interest difference
$1,489.57 → actual FRA payoff ✅

Therefore:

C. $1,489.57 ✅

what is Convenience yield ?

Convenience Yield — CFA Level I Derivatives

Convenience yield is the **non-monetary benefit of physically holding a commodity rather than holding a futures contract on that commodity**.

In simple words:

> **Convenience yield = the benefit you get from actually having the commodity available in inventory.**

Example: Oil

Imagine a refinery needs crude oil to operate.

It has two choices:

1. **Actually hold oil in storage**, or
2. Buy an **oil futures contract** and receive the oil later.

Why might the refinery want to physically hold oil?

Because having oil available immediately means it can:

- Keep production running even if supply is disrupted.
- Avoid shortages.
- Respond quickly to unexpected demand.
- Avoid the risk of not being able to obtain oil when needed.

That benefit is the **convenience yield**.

🔑 Convenience yield is NOT cash income

This is important.

If you hold a stock, you may receive:

> **Dividend → monetary benefit**

If you hold a commodity, you don't normally receive a dividend.

Instead, you may receive:

> **Convenience yield → non-monetary benefit**

So:

Asset	Benefit from holding it
Stock	Dividend
Bond	Coupon
Commodity	**Convenience yield**

Connection to commodity futures pricing

This is where CFA usually tests it.

For a commodity:

$$\boxed{F_0 = S_0 e^{(r+u-y)T}}$$

where:

- F_0 = futures price
- S_0 = spot price
- r = financing/carrying cost
- u = storage cost
- y = **convenience yield**

Notice that convenience yield has a **negative sign**.

Why?

Because convenience yield is a **benefit from holding the physical commodity**, so it **reduces the effective cost of carrying the commodity**.

Think of it as:

$$\boxed{\text{Net cost of carry} = \text{Financing cost} + \text{Storage cost} - \text{Convenience yield}}$$

So if convenience yield **increases**:

$$\boxed{\text{Futures price decreases}}$$

all else equal.

🧠 A very useful intuition

Imagine a commodity is **extremely scarce**.

Physical inventory becomes very valuable.

Companies desperately want to **have the commodity on hand**.

Therefore:

$$\text{Scarcity} \uparrow$$

$$\text{Value of holding inventory} \uparrow$$

$$\boxed{\text{Convenience yield} \uparrow}$$

This is why convenience yield is often **high when inventories are low**.

⭐ CFA Memory Trick

Remember:

> **Convenience yield = "I want the commodity NOW."**

If having the physical commodity provides a special benefit, that benefit is the **convenience yield**.

And for pricing:

$$\boxed{F = S + \text{costs} - \text{benefits}}$$

Therefore:

Storage + financing costs → increase futures price

Convenience yield → decreases futures price

A swap that involves the exchange of a fixed payment for a floating payment is most likely equivalent to a series of: A off-market forward contracts. B forward contracts that all have an initial positive value. C forward contracts that all have an initial value equal to the fixed payment.

✅ Correct answer: A. off-market forward contracts

This question is testing the relationship between a **swap and a series of forward contracts**.

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1. First, understand the basic idea

A **swap** involves exchanging cash flows on multiple future dates.

For example, an interest rate swap might involve:

> **Pay fixed rate ↔ Receive floating rate**

Suppose the swap has payments every 6 months for 3 years.

That's essentially:

- Forward contract #1 → settlement in 6 months
- Forward contract #2 → settlement in 12 months
- Forward contract #3 → settlement in 18 months
- ...
- Forward contract #6 → settlement in 36 months

So:

$$\boxed{\text{Swap} \approx \text{Series of forward contracts}}$$

---

2. Why are they "off-market" forwards?

This is the key.

At the **start of a swap**, its value is normally:

$$\boxed{0}$$

Neither party pays an upfront amount to enter a standard swap.

Therefore, the individual forward contracts making up the swap are structured so that their values **offset appropriately** to produce a zero-value package.

These are referred to as **off-market forwards**.

Therefore:

$$\boxed{\text{Swap} = \text{series of off-market forward contracts}}$$

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3. Why B is wrong?

B. Forward contracts that all have an initial positive value ❌

A standard swap is entered into at approximately **zero value**.

If every forward contract had a **positive initial value**, one party would be receiving value upfront.

That wouldn't describe a standard zero-value swap.

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4. Why C is wrong?

C. Forward contracts that all have an initial value equal to the fixed payment ❌

The fixed payment is a **cash flow exchanged at each settlement date**.

It is not the initial value of each forward contract.

Remember:

> **Fixed payment ≠ initial value of forward contract**

🧠 CFA Exam Shortcut

When you see:

> **Swap = series of ?**

Immediately think:

$$\boxed{\text{Swap} = \text{series of off-market forward contracts}}$$

Why "off-market"?

Because the individual forwards are structured around the swap's agreed fixed/floating terms rather than being newly initiated at prevailing market-forward terms.

Final answer:

$$\boxed{\textbf{A. off-market forward contracts}}$$

Memory rule:
Swap = Bundle/series of forwards → Off-market forwards → Initial swap value ≈ 0.

can you help me with example ?

Absolutely. Let's use a **very simple numerical example** to see why a swap can be viewed as a **series of forward contracts**.

Example: 2-year interest rate swap

Suppose Company A enters a 2-year swap with a bank.

- Notional = **$1,000,000**
- Payments every year
- Company A **pays fixed 5%**
- Company A **receives floating rate**

So each year, the cash flow is:

$$\text{Net payment} = (5\% - \text{Floating rate}) \times \$1,000,000$$

Assume the floating rates turn out to be:

Year	Fixed	Floating	Net cash flow
1	5%	4%	Company A pays $10,000
2	5%	6%	Company A receives $10,000

Now break the swap into forwards

Instead of thinking about **one 2-year swap**, think about **two separate forward contracts**:

Forward #1 — settles in Year 1

The agreement determines the difference between:

$$5\% \quad \text{and} \quad \text{Year-1 floating rate}$$

Floating = 4%:

$$(5\% - 4\%) \times 1{,}000{,}000 = \$10{,}000$$

Company A pays $10,000.

Forward #2 — settles in Year 2

The second forward determines the difference between:

$$5\% \quad \text{and} \quad \text{Year-2 floating rate}$$

Floating = 6%:

$$(5\% - 6\%) \times 1{,}000{,}000 = -\$10{,}000$$

Negative from A's perspective means **A receives $10,000**.

So look at the big picture

Instead of:

> **One 2-year swap**

we can think of:

> **Forward #1 + Forward #2**

Each forward corresponds to **one future settlement/payment period**.

Therefore:

$$\boxed{\text{Interest Rate Swap} \approx \text{Series of Forward Contracts}}$$

But why "off-market"?

Here's the important CFA distinction.

Imagine today's market forward rates are:

- Year 1 forward rate = 4%
- Year 2 forward rate = 6%

But the swap has a fixed rate of **5%**.

The swap combines forward-like agreements using the **5% swap rate**.

The individual forwards therefore aren't necessarily ordinary **at-market forwards** with zero value when viewed separately.

They are called **off-market forwards**.

🧠 Think of the structure like this:

SWAP

⬇️

Forward 1 → payment at Year 1
Forward 2 → payment at Year 2
Forward 3 → payment at Year 3
...

Each forward compares:

$$\boxed{\text{Fixed rate vs. Floating rate}}$$

And together they create the swap's cash flows.

⭐ CFA memory rule

> **Swap = series of off-market forwards**

And:

> **One swap payment period ≈ one forward contract.**

That's why **A** was the correct answer in your previous question.

For a stock that pays no dividends, the value of an American call option is most likely: A the same as the value of a European call option with otherwise identical features. B greater than the value of a European call option with otherwise identical features. C less than the value of a European call option with otherwise identical features.

✅ Correct answer: A. The same as the value of a European call option

This is a **very important CFA Level I rule**.

The question says:

> **Stock pays no dividends**

That is the key.

---

1. American vs. European options

The difference is **when you can exercise**:

- **American call** → can exercise **any time before or at expiration**
- **European call** → can exercise **only at expiration**

Normally, because an American option gives you **more flexibility**, we would expect:

$$C_{American} \geq C_{European}$$

But for a **non-dividend-paying stock**, something special happens.

---

2. Why wouldn't you exercise an American call early?

Suppose:

- Stock price = $100
- Strike price = $80
- Time remaining = 6 months

Your American call is already **in the money**.

You might think:

> "Why not exercise now and get the $20 profit?"

But exercising early means you pay the **$80 strike price now**.

If you **don't exercise**, you can keep the call and wait.

You retain:

① Time value

The stock could rise further.

② Use of the strike price

You don't have to pay the $80 until expiration.

That means you can keep/invest that money until then.

So **early exercise provides no advantage** when the stock pays no dividends.

---

3. Therefore...

Because there is **never an optimal reason to exercise a call early** on a non-dividend-paying stock:

$$\boxed{C_A = C_E}$$

where:

- C_A = American call value
- C_E = European call value

Therefore:

✅ A. Same as the value of a European call

🧠 CFA Exam Rule — VERY IMPORTANT

Memorize this:

Call on non-dividend-paying stock

$$\boxed{\text{American Call} = \text{European Call}}$$

Because **early exercise is never optimal**.

What about a dividend-paying stock?

That's where things change.

If the stock pays a **dividend**, exercising the call before the ex-dividend date can sometimes be beneficial because the **call holder does not receive the dividend**, while the stockholder does.

Therefore:

$$\boxed{\text{Dividend-paying stock} \rightarrow \text{American call can be worth more}}$$

Quick memory table

Situation	Relationship
No dividends + Call	🇺🇸 American Call = European Call
Dividends + Call	American Call ≥ European Call
Any put	American Put ≥ European Put

⭐ Exam shortcut

When you see:

> **"American call + stock pays no dividends"**

Immediately think:

$$\boxed{\textbf{SAME}}$$

Answer: A ✅

can you help this case : What about a dividend-paying stock? with proper example?

Yes. This is a very important distinction in CFA Derivatives. The key is understanding **why dividends can make early exercise of an American call worthwhile**.

---

1. First, imagine a stock that pays NO dividend

Suppose:

- Stock = $100
- Call strike = $80
- 3 months remaining

You have an **American call**.

You could exercise now:

$$100 - 80 = \$20$$

But why would you?

If you don't exercise, you keep the call and retain its **time value**. You also don't have to pay the $80 strike until you exercise.

So there is generally **no reason to exercise early**.

Therefore:

$$\boxed{\text{American Call} = \text{European Call}}$$

---

2. Now introduce a dividend

Suppose the stock is:

- Current stock price = **$100**
- Strike price = **$90**
- American call
- Dividend = **$5**
- Dividend is going to be paid tomorrow
- Assume the call expires later.

Here's the important thing:

If you own the stock:

You receive the **$5 dividend**.

If you own the call:

You **do NOT receive the dividend**.

This creates a reason to exercise the call **before the stock goes ex-dividend**.

---

3. Let's see what happens

You own a call:

$$X = \$90$$

Stock:

$$S = \$100$$

The call is in the money by:

$$100 - 90 = \$10$$

Suppose you exercise **before the dividend**.

You pay:

$$\$90$$

and receive the stock worth:

$$\$100$$

Then you own the stock and receive the upcoming:

$$\boxed{\$5 \text{ dividend}}$$

So your immediate economic benefit is approximately:

$$\$10 + \$5 = \boxed{\$15}$$

The $5 dividend gives you an **additional benefit from exercising early**.

---

4. What happens if you DON'T exercise?

If you continue holding the call:

- You keep the option's time value.
- But you **don't receive the $5 dividend**.

Then, when the stock goes ex-dividend, its price will typically fall by approximately the dividend amount, all else equal.

For example:

Before dividend:

$$S = \$100$$

Dividend:

$$\$5$$

After dividend:

$$S \approx \$95$$

Your call becomes less valuable because the underlying stock price has fallen.

---

5. So the investor faces a trade-off

This is the key concept.

Exercise early:

You:

✅ Get the stock
✅ Receive the dividend
❌ Give up the remaining time value of the call
❌ Pay the strike price earlier

Don't exercise:

You:

✅ Keep the call's remaining time value
✅ Delay paying the strike price
❌ Don't receive the dividend
❌ Stock price may fall when dividend is paid

Therefore, **if the dividend is sufficiently large**, early exercise can be optimal.

---

6. This is why the CFA rule changes

Non-dividend-paying stock:

$$\boxed{\text{American Call} = \text{European Call}}$$

because early exercise is not optimal.

Dividend-paying stock:

$$\boxed{\text{American Call} \geq \text{European Call}}$$

and **it can be optimal to exercise early**.

Therefore:

$$\boxed{\text{American Call can be worth MORE than European Call}}$$

🧠 The intuition I want you to remember

Think of a call holder as someone who has **the right to buy the stock**, but doesn't own the stock yet.

Therefore:

> **Call holder does NOT receive dividends.**

But if you exercise the call:

> **You become the stockholder → now you receive the dividend.**

That's the entire reason dividends matter for early exercise.

⭐ CFA Exam Memory Rule

Call + No dividend:

Don't exercise early.

$$\boxed{A_C = E_C}$$

Call + Dividend:

Early exercise MAY be optimal, particularly just before a sufficiently large dividend.

$$\boxed{A_C \geq E_C}$$

Put:

An American put can also be exercised early, even without dividends, because exercising lets you receive the strike price earlier.

$$\boxed{A_P \geq E_P}$$

One-line memory trick

> **Dividend gives a reason to exercise a CALL early because exercising converts you from a call holder into a stockholder who receives the dividend.**

Which of the following attributes is least likely to be a requirement for the existence of riskless arbitrage? The underlying security: A can be sold short. B is a financial asset. C is relatively liquid.

Correct answer: B. is a financial asset

The key idea is that **riskless arbitrage requires the ability to exploit a price difference** between equivalent positions. For that, certain characteristics of the underlying security are important.

Let's break down each choice.

A. Can be sold short → Requirement

Short selling is important for arbitrage because it allows you to profit when an asset is **overpriced**.

For example:

- Stock A should be worth ₹100
- But it is trading at ₹110

An arbitrageur can **short sell** it at ₹110 and simultaneously buy an equivalent position elsewhere at ₹100.

The ₹10 difference is the arbitrage profit.

So, the ability to short sell is generally an important condition for riskless arbitrage.

B. Is a financial asset → NOT a requirement ✅

The underlying does **not necessarily have to be a financial asset**.

Arbitrage can exist in:

- Commodities
- Currencies
- Financial securities
- Physical goods

For example, if gold is simultaneously trading at different prices in two markets, an arbitrageur can potentially buy in the cheaper market and sell in the more expensive market.

Therefore, being a **financial asset is not essential**.

C. Is relatively liquid → Requirement

Liquidity is important because the arbitrageur needs to be able to:

1. Buy the cheaper asset.
2. Sell the more expensive asset.
3. Execute both transactions quickly.
4. Do so without significant transaction costs or price movement.

If the asset is highly illiquid, the apparent price difference may disappear before the trades can be executed.

So relative liquidity supports the existence of practical riskless arbitrage.

CFA Exam Shortcut 🧠

For **riskless arbitrage**, think:

> **Can I buy cheap + sell expensive + execute both trades?**

Therefore, you want:

Attribute	Required?	Why?
Can be sold short	✅	Allows arbitrage against an overpriced asset
Financial asset	❌	Arbitrage can involve physical/commodity assets

Attribute	Required?	Why?
Relatively liquid	✅	Allows simultaneous execution

Answer: B. is a financial asset ✅

Memory rule:

> **Arbitrage cares about tradability, shortability, and liquidity—not whether the asset is classified as a financial asset.**

According to put–call parity, if a fiduciary call expires in the money, the payoff is most likely equal to the: A difference between the market value of the asset and the face value of the risk-free bond. B market value of the asset. 2020 Level I Mock Exam (A) PM 55 C face value of the risk-free bond.

Correct answer: B. Market value of the asset ✅

This question is testing your understanding of a **fiduciary call** and **put–call parity**.

Let's build it from scratch.

---

1. What is a fiduciary call?

A **fiduciary call** is:

> **Long call + risk-free zero-coupon bond**

The bond is purchased so that its **face value equals the exercise price** of the call.

So:

$$\boxed{\text{Fiduciary Call} = \text{Long Call} + \text{Risk-free Bond}}$$

The bond will provide exactly **X (exercise price)** at expiration.

---

2. What happens if the call expires ITM?

Suppose:

- Current stock price = $100
- Exercise price X = $80
- Risk-free bond face value = $80
- At expiration, stock price = $120

Because:

$$S_T > X$$

the call is **in the money**.

Call payoff:

$$S_T - X = 120 - 80 = 40$$

Bond payoff:

$$X = 80$$

Therefore, the fiduciary call payoff is:

$$40 + 80 = \boxed{120}$$

And what is $120?

👉 **The market value of the underlying asset.**

Why?

When the call is exercised:

Call gives you the stock for $80.

The bond gives you exactly **$80 cash**.

You use the $80 from the bond to pay the exercise price and receive the stock.

So ultimately, you own the **entire stock**, worth $120.

At expiration:

$$(S_T - X) + X = S_T$$

Therefore:

$$\boxed{\text{Fiduciary call payoff} = S_T}$$

when the call expires **in the money**.

Why the other answers are wrong

A. Difference between market value of asset and face value of bond ❌

That is only the **call's payoff**:

$$S_T - X$$

It does **not** include the bond payoff.

The question asks about the **fiduciary call**, which contains both the call and the bond.

B. Market value of the asset ✅

Correct.

$$\boxed{(S_T - X) + X = S_T}$$

The call gives you the stock's value above the strike, while the bond provides the strike amount.

C. Face value of the risk-free bond ❌

The bond's face value is only:

$$X$$

But the fiduciary call consists of **both** the call and the bond.

So its payoff is greater than X when the call is ITM.

🧠 CFA Memory Rule

Remember the two portfolios from put–call parity:

Portfolio	Components	Payoff at expiration
Fiduciary call	Long call + bond	S_T if call ITM

Portfolio	Components	Payoff at expiration
Protective put	Long stock + long put	X if stock is below X

The most important identity:

$$\boxed{\text{Fiduciary call} = \text{Protective put}}$$

by put–call parity.

Super-short exam shortcut:

> **Fiduciary call = call + cash → if ITM, cash pays the strike and call gets the stock → total = stock value.**

Therefore, **B. market value of the asset**.

can you better help me understand the concept with example and equation ?

Absolutely. The easiest way to understand a **fiduciary call** is to forget put–call parity for a moment and just understand **what the two pieces do at expiration**.

1. First: What is a fiduciary call?

A fiduciary call is a portfolio containing:

$$\boxed{\text{Long Call} + \text{Risk-free zero-coupon bond}}$$

The bond has a **face value equal to the call's exercise price** X.

So:

$$\boxed{\text{Fiduciary Call} = C + PV(X)}$$

where:

- C = call option
- X = exercise price
- $PV(X)$ = present value of the exercise price
- At expiration, the bond becomes worth X.

---

2. Let's use a very simple example

Suppose:

- Stock price today = **$100**
- Call exercise price = **$80**
- Call expires in 1 year
- Risk-free bond has a **face value of $80**

So our fiduciary call is:

$$\boxed{\text{Call} + \$80 \text{ bond}}$$

Now let's see what happens at expiration.

Scenario 1: Stock finishes at $120

The stock is above the exercise price:

$$S_T = 120 > X = 80$$

Therefore, the call is **in the money**.

Step 1: What does the call give us?

A call gives us the right to buy the stock for $80.

The stock is worth $120.

Therefore, the call's payoff is:

$$\text{Call payoff} = S_T - X$$

$$= 120 - 80$$

$$= \boxed{\$40}$$

Step 2: What does the bond give us?

The bond has a face value of $80.

At expiration:

$$\boxed{\text{Bond payoff} = \$80}$$

Step 3: Combine them

Fiduciary call payoff:

$$\text{Call payoff} + \text{Bond payoff}$$

$$= 40 + 80$$

$$= \boxed{\$120}$$

And notice:

$$\$120 = S_T$$

So:

$$\boxed{\text{Fiduciary call payoff} = S_T}$$

That's why the answer is **B. market value of the asset**.

---

3. But WHY does this happen?

This is the most important part.

You have:

Call

The call gives you the **stock's value above $80**:

$$120 - 80 = \$40$$

Bond

The bond gives you the **$80 needed to pay the exercise price**:

$$\$80$$

Put them together:

$$\underbrace{40}_{\text{call}} + \underbrace{80}_{\text{bond}} = 120$$

So you've effectively obtained the **entire stock**.

Think of it this way:

> **Call gives me the stock's upside above X.**
> **Bond gives me X.**
> **Together, they give me the whole stock.**

---

4. The equation behind it

For a call:

$$\boxed{\text{Call payoff} = \max(S_T - X, 0)}$$

For the bond:

$$\boxed{\text{Bond payoff} = X}$$

Therefore:

$$\text{Fiduciary call payoff} = \max(S_T - X, 0) + X$$

If the call is **ITM**, then:

$$S_T > X$$

so:

$$\max(S_T - X, 0) = S_T - X$$

Therefore:

$$\text{Fiduciary call payoff} = (S_T - X) + X$$

The X's cancel:

$$\boxed{\text{Fiduciary call payoff} = S_T}$$

That's the entire logic behind the answer.

---

5. What if the call expires OUT of the money?

This is also important for understanding the concept.

Suppose instead:

$$S_T = \$60$$

and:

$$X = \$80$$

The call is worthless because:

$$60 < 80$$

Call payoff:

$$\max(60 - 80, 0) = 0$$

Bond payoff:

$$80$$

Therefore:

$$\boxed{\text{Fiduciary call payoff} = \$80}$$

So we can summarize:

Stock at expiration	Call payoff	Bond payoff	Fiduciary call payoff
$60	$0	$80	**$80**
$80	$0	$80	**$80**
$100	$20	$80	**$100**
$120	$40	$80	**$120**

Notice the pattern:

$$\boxed{\text{Fiduciary call payoff} = \max(S_T, X)}$$

Because:

$$\max(S_T - X, 0) + X = \max(S_T, X)$$

So:

If $S_T > X$:

$$\boxed{\text{Fiduciary call payoff} = S_T}$$

If $S_T < X$:

$$\boxed{\text{Fiduciary call payoff} = X}$$

---

6. Now connect it to put–call parity

This is where CFA questions become easier.

Put–call parity says:

$$\boxed{C + PV(X) = P + S_0}$$

Let's identify the two sides.

Left side:

$$C + PV(X)$$

This is:

$$\boxed{\text{Fiduciary call}}$$

because:

> Call + risk-free bond = fiduciary call.

Right side:

$$P + S_0$$

This is:

$$\boxed{\text{Protective put}}$$

because:

> Stock + put = protective put.

Therefore:

$$\boxed{\text{Fiduciary call} = \text{Protective put}}$$

---

7. Why are these two portfolios equivalent?

Let's use the same $100 stock, $80 strike example.

Portfolio A: Fiduciary call

$$\boxed{\text{Call} + \text{Bond}}$$

Portfolio B: Protective put

$$\boxed{\text{Stock} + \text{Put}}$$

Now consider two possible outcomes.

If stock = $120

Fiduciary call:

$$40 + 80 = 120$$

Protective put:

Stock = $120
Put = $0

$$120 + 0 = 120$$

Same payoff.

If stock = $60

Fiduciary call:

Call = $0
Bond = $80

$$0 + 80 = 80$$

Protective put:

Stock = $60
Put payoff:

$$80 - 60 = 20$$

Therefore:

$$60 + 20 = 80$$

Again:

$$\boxed{80 = 80}$$

So both portfolios have exactly the same payoff.

🧠 The big picture

Think of the fiduciary call as a **stock-replication strategy** when the call finishes ITM.

Fiduciary call:

$$\boxed{\text{Call} + \text{Bond}}$$

Call gives:

$$S_T - X$$

Bond gives:

$$X$$

Together:

$$(S_T - X) + X = S_T$$

Therefore:

$$\boxed{\text{If call is ITM} \rightarrow \text{fiduciary call payoff} = \text{stock value}}$$

⭐ CFA Memory Rule

Memorize these three equations:

$$\boxed{\text{Fiduciary Call} = C + PV(X)}$$

$$\boxed{\text{Protective Put} = S + P}$$

$$\boxed{C + PV(X) = S + P}$$

And for the specific question:

> **If the fiduciary call expires ITM, the call gives $S_T - X$, while the bond gives X. Together they equal S_T.**

Therefore:

$$\boxed{\text{Answer} = \text{B. Market value of the asset}}$$

One-line intuition:
Call gives you the part of the stock above the strike; the bond gives you the strike; together you have the whole stock.

Which statement best describes the early exercise of non-dividend paying American options? Early exercise may be advantageous for: A both deep-in-the-money calls and deep-in-the-money puts. B deep-in-the-money calls. C deep-in-the-money puts.

Correct answer: C. Deep-in-the-money puts ✅

This is a very important CFA rule. Let's understand **why a put can be exercised early but a call generally cannot** when the stock pays **no dividends**.

---

1. First, what does "early exercise" mean?

An **American option** can be exercised **any time before expiration**.

A **European option** can only be exercised **at expiration**.

The question asks:

> For a **non-dividend-paying stock**, which option might benefit from being exercised early?

Answer:

$$\boxed{\text{Deep-in-the-money PUT}}$$

---

2. Why NOT a call?

Suppose:

- Stock price = $50
- Call strike = $40
- Call is deep ITM
- Stock pays **no dividends**

The call allows you to buy the stock for $40 when it is worth $50.

So intrinsic value is:

$$S - X = 50 - 40 = \$10$$

You might think:

> "Why not exercise now and make $10?"

Because by exercising early, you **give up the remaining time value of the call**.

More importantly, you have to pay the $40 **today** instead of waiting until expiration.

If you simply hold the call:

- You keep the option's time value.
- You don't have to pay $40 yet.
- Your $40 can remain invested and earn interest.

So early exercise of a call on a **non-dividend-paying stock is never optimal**.

Therefore:

$$\boxed{\text{No dividends} + \text{Call} \rightarrow \text{Don't exercise early}}$$

---

3. Why can a put be exercised early?

Now consider a put.

Suppose:

- Stock price = $20
- Put strike = $100
- Put is **deep ITM**

The put allows you to sell the stock for $100 even though it's worth only $20.

Put intrinsic value:

$$X - S = 100 - 20$$

$$= \$80$$

If you exercise now, you receive **$100**.

That's attractive because you get the $100 **today** rather than waiting until expiration.

You can then invest that $100 and earn interest.

---

4. The key difference

This is the intuition CFA wants you to understand.

Call

Exercising early means:

> **You PAY the strike price early.**

That's bad because you're giving up the ability to keep your money invested.

Put

Exercising early means:

> **You RECEIVE the strike price early.**

That's potentially good because you can invest the money and earn interest.

---

5. Numerical example

Suppose:

- Strike price = $100
- Stock price = $20
- One year remaining
- Risk-free rate = 5%

If you exercise the put today:

You receive:

$$\$100$$

You can invest that $100 for one year:

$$100(1.05) = \$105$$

That's attractive.

If you wait until expiration:

You don't receive the $100 until later.

So receiving the strike price **earlier** has value.

---

6. Why "deep-in-the-money" matters

A slightly ITM put may still have significant **time value**.

For example:

$$S = 95, \quad X = 100$$

Intrinsic value:

$$100 - 95 = \$5$$

There may be enough time value remaining that holding the put is better than exercising.

But if the put is **deep ITM**:

$$S = 20, \quad X = 100$$

Intrinsic value is:

$$100 - 20 = \$80$$

The option is heavily ITM, and the benefit of receiving the $100 early can outweigh the remaining time value.

Therefore, early exercise **may be advantageous**.

---

7. CFA rule to memorize 🧠

Option	Non-dividend-paying stock	Early exercise?
American Call	No dividends	❌ **Never optimal**
American Put	No dividends	✅ **May be optimal**
Deep ITM Put	No dividends	⭐ **Most likely**

The golden rule:

$$\boxed{\text{Non-dividend stock} + \text{Call} \rightarrow \text{Never exercise early}}$$

$$\boxed{\text{Non-dividend stock} + \text{Put} \rightarrow \text{Early exercise may be optimal}}$$

And because the question specifically asks **"may be advantageous for"**, the best answer is:

$$\boxed{\textbf{C. Deep-in-the-money puts}}$$

One-second CFA shortcut:

> **Call → paying strike early = bad.**
> **Put → receiving strike early = potentially good.**

ChatGPT can make mistakes. Check important info.